Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/23/2005
PI Name:	Chapman, Michael Ph.D.		
Project Title:	All -Optical Atomic Bose Condensates		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30332-0430	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	04/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2893		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We propose an experimental program to investigate atomic Bose-Einstein condensates created using all-optical trapping methods. We will exploit the considerable flexibility provided by tailoring the optical trapping potential to explore alternative trapping geometries, allowing for for-mation of multiple condensates. This capability, combined with spin-independent nature of these traps, will allow us to explore many interesting fundamental issues of multi-component conden-sates. Our studies will include detailed examinations of the remarkable loading properties of these traps. The objective of our program is to investigate Bose-Einstein condensates formed by all-optical techniques. We will focus on several unique aspects of our system, including formation dynamics of our multi-component condensates, loading dynamics of our traps, and creation and study of condensates in multiple traps. In particular, we will further develop our all-optical BEC approach, endeavoring to create larger condensates by		

	tailoring the optical potentials. Additionally, we will apply these techniques to the study of multicomponent (ferromagnetic) spinor condensates and interacting condensates in large period optical lattices. Atomic Bose condensates may form the basis for next generation atomic clocks in space. Space-based ultracold atomic experiments may benefit significantly from the developments resulting from our program, particularly with regards to the speed and simplicity of the condensate formation together with the low power consumption of our trapping lasers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have developed a new technique to create large atomic Bose-Einstein condensates in a single focus optical trap. We initially load 30 million atoms into the optical trap. After 2 sec of rapid evaporation and dynamic compression, 300,000 atoms are cooled to Bose condensation. This new method not only maintains the simplicity and speed of our previous all-optical technique, but it also creates condensates 10x larger than before and is ideal for study BEC spin dynamics in a quasi 1-D system.
Bibliography Type:	Description: (Last Updated: 01/23/2008)
Articles in Peer-reviewed Journals	Chang MS, Hamley CD, Barrett MD, Sauer JA, Fortier KM, Zhang W, You L, Chapman MS. "Observation of spinor dynamics in optically trapped 87Rb Bose-Einstein condensates." Phys Rev Lett. 2004 Apr 9;92(14):140403. PMID: 15089521 , Apr-2004